

Synoptic LE: Tree Fluxes 2022–2023

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1 Read in and collate all the files in the Processed Data folder

2 Remove bad fluxes and calculate how many there were

```
##Remove any samples with "remove" in the qaqc_flag columns
# stem_flux_2022_2023 <- stem_flux_2022_2023 %>%
#   filter(!grepl("remove", ch4_qaqc_flag, ignore.case = TRUE)) %>%
#   filter(!grepl("remove", co2_qaqc_flag, ignore.case = TRUE))

##Removed fluxes in the CH4 and CO2 flux processing Rmd code, removed both CH4 and CO2 obs w
```

```

r2_summary <- stem_flux_2022_2023 %>%
  mutate(neg_C02 = co2_lin_flux.estimate < 0 | is.na(co2_lin_flux.estimate),
  summarise(total_obs = n(),
            n_neg_C02 = sum(neg_C02, na.rm = TRUE),
            pct_neg_C02 = 100 * n_neg_C02 / total_obs)

r2_summary

```

	total_obs	n_neg_C02	pct_neg_C02
1	424	13	3.066038

```

stem_flux_2022_2023 <-stem_flux_2022_2023%>%
  filter(co2_lin_flux.estimate>0)

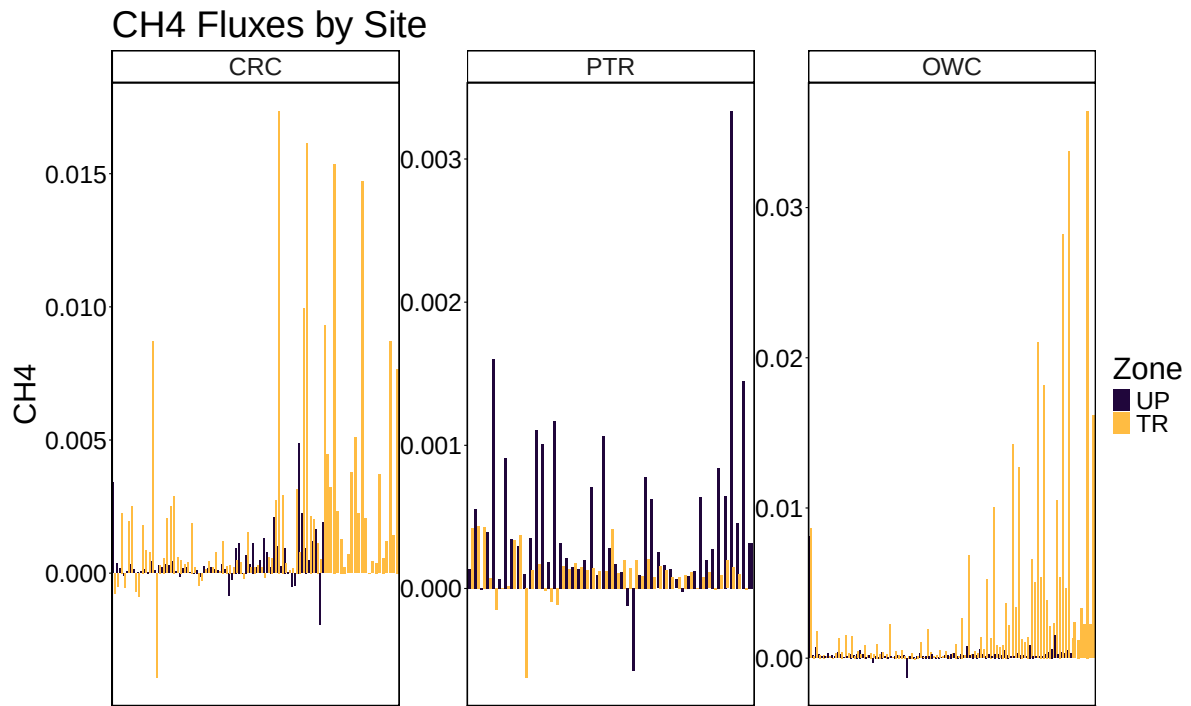
##note that this list has some not synoptic monitoring samples zone-labeled as "Luci" that w

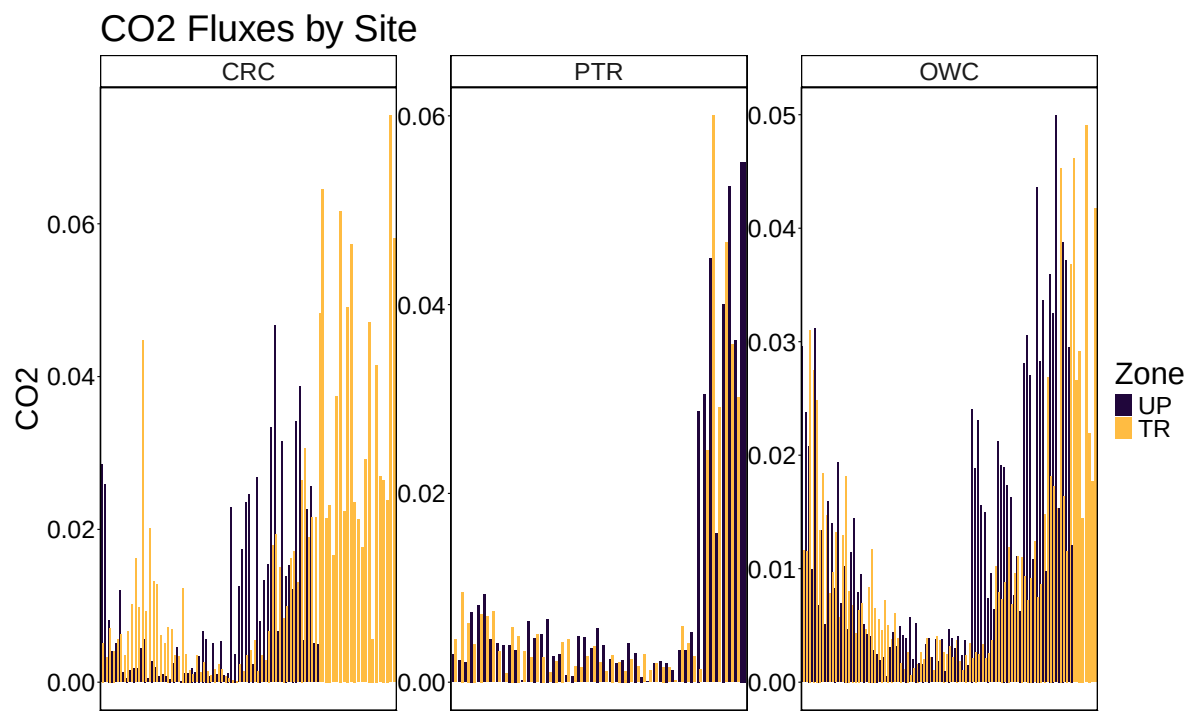
##note that this list has some diurnal variation samples for PTR synoptic veg campaign in 20

```

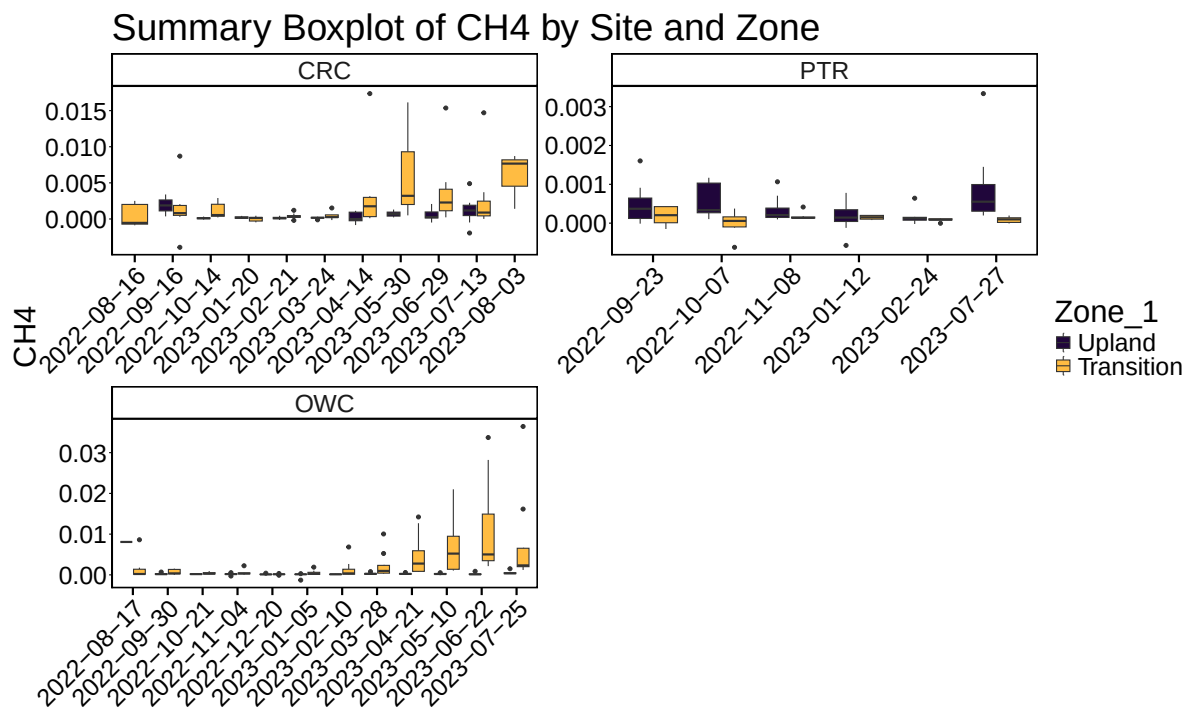
3 Write out files

4 All Data by Year, Site and Zone

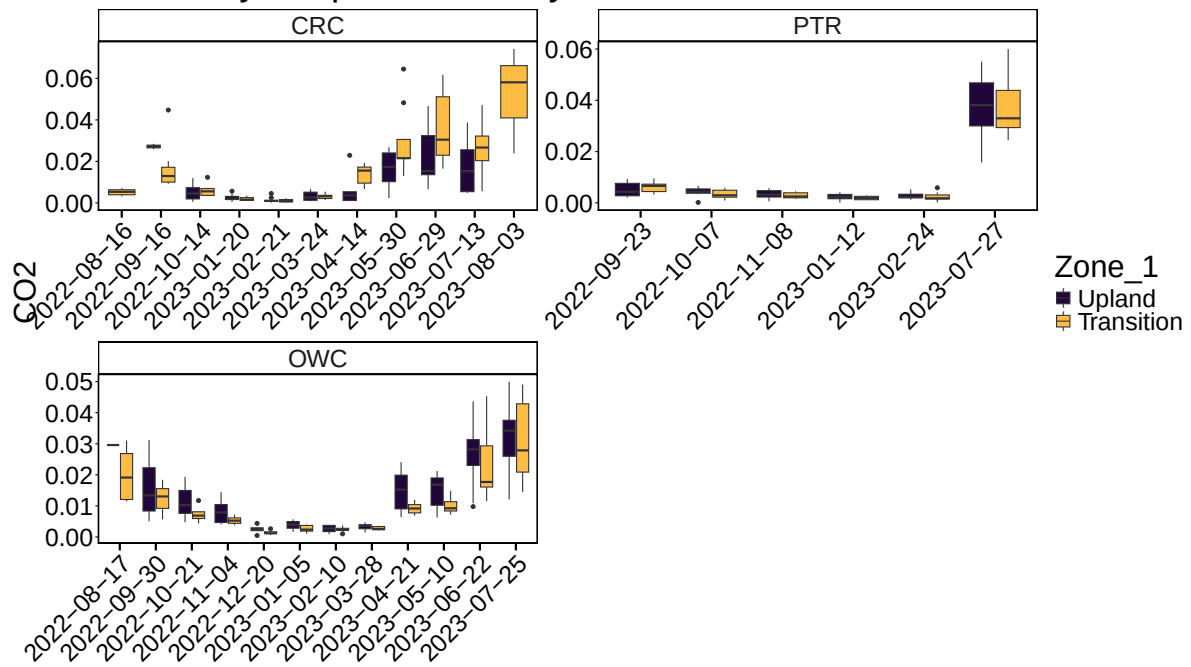




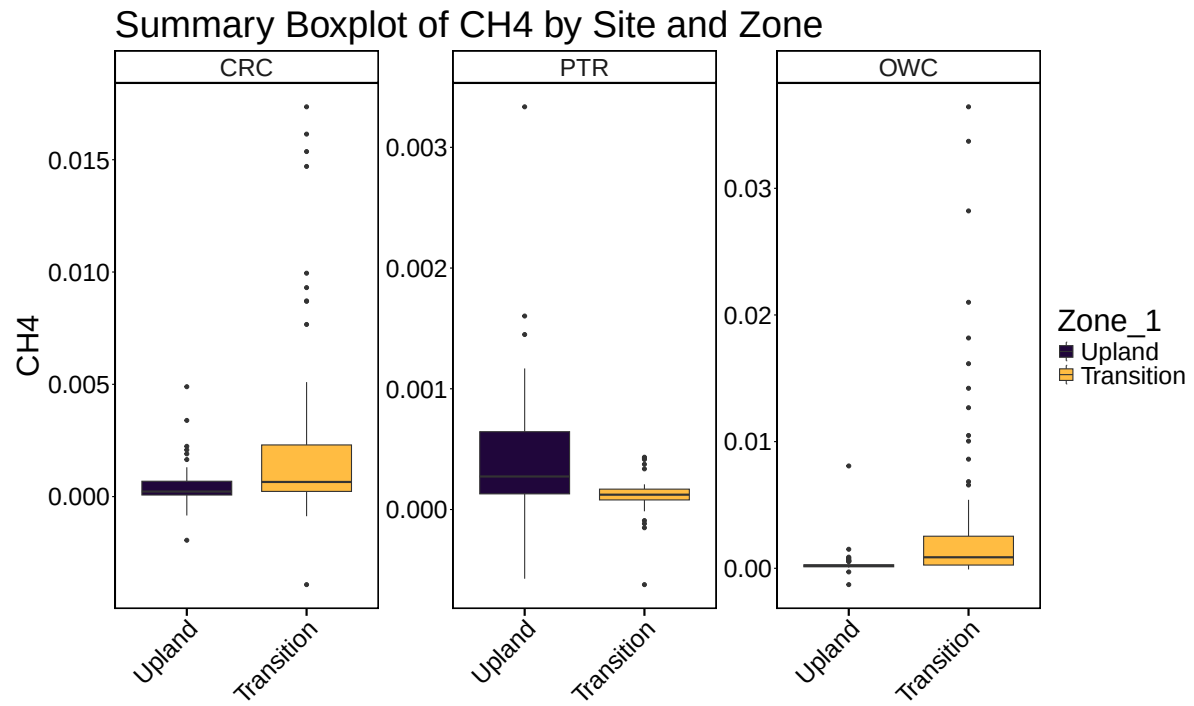
5 Boxplot summary by Date

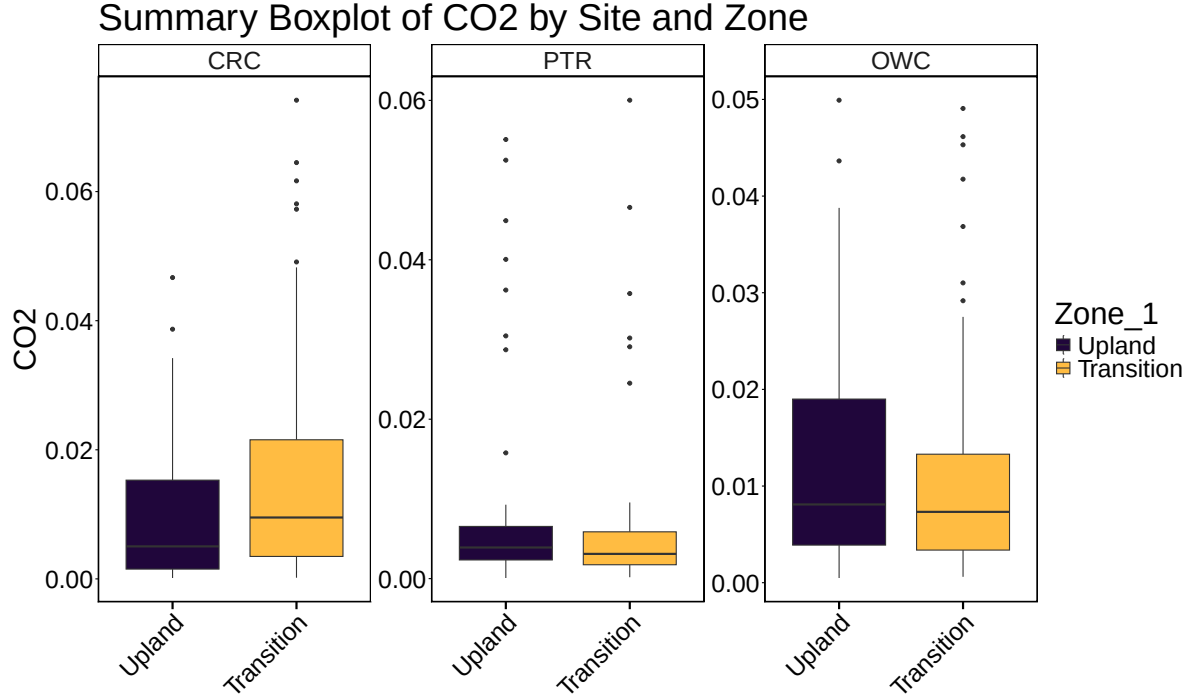


Summary Boxplot of CO2 by Site and Zone



6 Boxplot summary by Site





7 Calculate summary metrics

8 Summary Tables

Table 1: Summary Statistics by Site and Zone

Site	Zone	Min CH4	Max CH4	Mean CH4	Min CO2	Max CO2	Mean CO2
CRC	Upland	-0.0019429	0.0048932	0.0004792	0.0001376	0.0466813	0.0101341
CRC	Transition	-0.0039193	0.0173559	0.0021846	0.0001799	0.0741587	0.0159742
PTR	Upland	-0.0005731	0.0033354	0.0004563	0.0000782	0.0550926	0.0094212
PTR	Transition	-0.0006221	0.0004333	0.0001159	0.0001767	0.0600224	0.0078175
OWC	Upland	-0.0012904	0.0080793	0.0003185	0.0004861	0.0499195	0.0127369
OWC	Transition	-0.0000933	0.0364326	0.0034623	0.0006097	0.0490653	0.0107509